

Sangre de grado *Croton palanostigma* induces apoptosis in human gastrointestinal cancer cells.

Sangre de grado is an ethnomedicinal red tree sap obtained from *Croton* spp. that is used to treat gastrointestinal ulcers, cancer and to promote wound healing. To evaluate the potential role of sangre de grado (SdG) in cancer we examined its effects on human cancer cells, AGS (stomach), HT29 and T84 (colon). Viability of cells treated with SdG (10-200 microg/ml) decreased ($P < 0.01$) in a dose dependent manner measured over a 24-h period. Cell proliferation at 48 h decreased ($P < 0.01$) in all cells treated with SdG (> 100 microg/ml). When cells in suspension were treated with SdG (100 microg/ml) cell adherence was severely compromised ($> 85\%$). Cells treated with SdG (100 microg/ml) underwent apoptosis as detected by nucleus condensation and DNA fragmentation determined by ELISA, and flow cytometry. Morphological changes as assessed by acridine orange. These effects were similar to that observed with Taxol (30 microM). A significant alteration of microtubular architecture was equally observed in both stomach and colon cancer cells exposed to SdG (100 microg/ml). The induction of apoptosis and microtubule damage in AGS, HT29 and T84 cells suggest that sangre de grado should be evaluated further as a potential source of anti-cancer agents.